THERMAL FAULT DIAGNOSTIC REPORT

Automated Infrared Thermographic Analysis & Deep Learning Evaluation

Analysis ID:	3dfb77cf-53da-44ce-8bfe-3cc17a2bec8d	**Timestamp:**	2026-10-14T13:54:23
Image Filename:	test_thermal_transformer.png	**Model Version:**	ResNet18-ThermalTransfer (v1.2.0-trained)
Condition:	**Fault Detected**	**Fault Class:**	**Abnormal Hotspot**
Confidence Score:	31.0%	**Severity Level:**	**Medium**

Thermographic Inspection Imagery

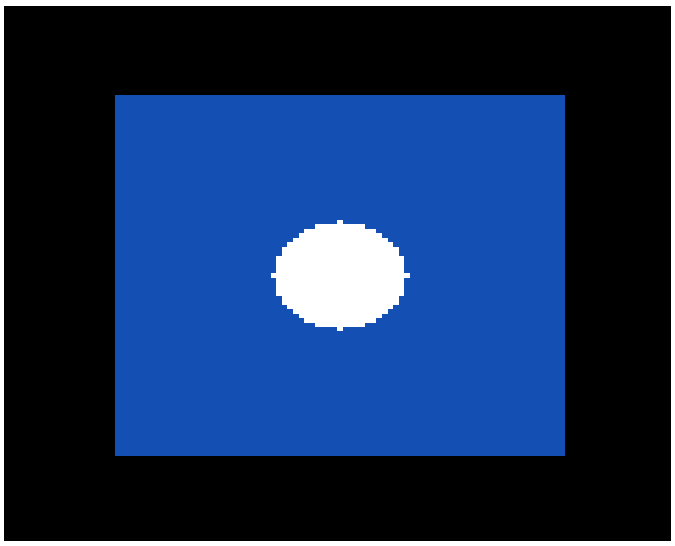

Figure 1: Original Thermal Image

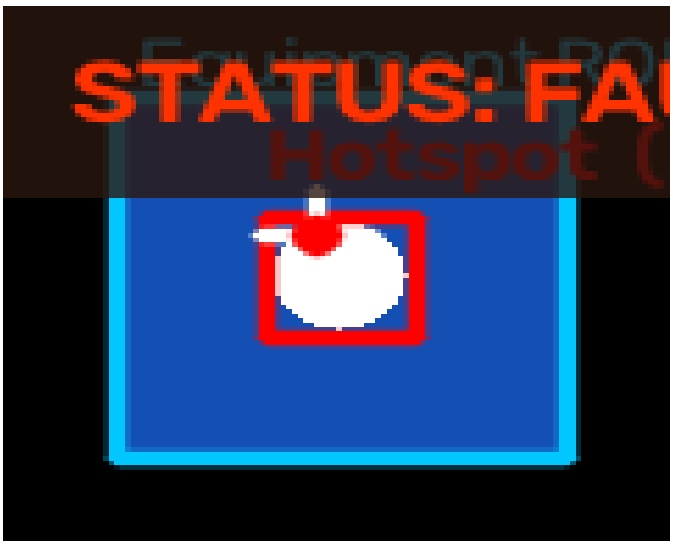

Figure 2: AI Diagnostic Overlay (ROI & Hotspot)

Hotspot Localization & Quantitative Metrics

Hotspot Bounding Box:	X: 47..74, Y: 47..74 (Width: 27px, Height: 27px)
Centroid Coordinates:	[60, 60] px
Peak Thermal Intensity Pixel:	[56, 51] px (Max Intensity: 255.0)
Relative Contrast / Area:	Contrast Ratio: +104.0% | Area Ratio: 8.35%

Engineering Interpretation & Recommendations

Summary: Moderate thermal anomaly identified (Abnormal Hotspot). Local hotspot exceeds equipment operating baseline by 104.0%. Schedule verification within standard maintenance cycle.

Next Actions: Schedule immediate physical inspection of identified terminal contacts. Verify torque on bolted electrical connections, inspect dielectric insulation, and measure phase current balance.

LIMITATIONS & DISCLAIMER:

• Severity levels are derived from thermal gradient contrast and geometry. Uncalibrated digital image formats (PNG/JPG) do not provide calibrated radiometric sensor data in degrees Celsius.

• Solar reflection, airflow, and surface emissivity affect infrared measurements.

• **Automated analysis supports inspection and should not replace qualified maintenance assessment.**